AI-Assisted Data Visualization for Business Decision Support Systems

An Integrated Framework for Exploratory Analytics, Profit Prediction, Explainable AI, Hybrid Model Evaluation, and Interactive Decision Support

Research Paper — Review 3 Draft

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Data Science and Visualization

ABSTRACT

This paper presents an AI-assisted business analytics and decision-support framework that integrates exploratory data analysis, feature engineering, hybrid feature selection, supervised profit prediction, explainable artificial intelligence, business recommendations, and interactive visualization. The implementation is designed for three heterogeneous retail/e-commerce datasets: Tableau Sample Superstore, Global Superstore, and the Brazilian E-Commerce Public Dataset (Olist). The datasets are intentionally analyzed according to their available schemas rather than being naively merged. Superstore and Global Superstore support transaction-level profit modelling, while the available Olist customer file supports customer-geographic segmentation. The methodology follows the faculty-prescribed research progression: study published models, reproduce baseline techniques, identify limitations, evaluate multiple hybrid alternatives, and select a final candidate only after validation and comparative testing. The current validated experiments include Naive, Ridge Regression, Random Forest, Gradient Boosting, feature-selection-aware Random Forest, and several hybrid candidates. On the latest Superstore test experiment, Gradient Boosting achieved MAE 21.7034, RMSE 93.7295, sMAPE 53.5164, and RZ 0.8369, while the strongest genuine hybrid candidate among the completed candidates, a simple RF+GB combination, achieved MAE 24.4975, RMSE 102.2760, sMAPE 57.0464, and RZ 0.8059. On Global Superstore, Gradient Boosting achieved MAE 35.5769, RMSE 94.1239, sMAPE 69.7579, and RZ 0.6671. These results show that a hybrid should not be claimed as superior merely because it is a hybrid; the final proposed model must demonstrate a measurable and defensible advantage against strong baselines. Accordingly, this paper distinguishes completed evidence from the final hybrid-selection experiment that remains to be executed. The framework also provides model-agnostic permutation importance, deterministic business recommendations, and a Streamlit dashboard for decision support. The principal contribution is therefore an integrated, evidence-traceable business analytics pipeline rather than a claim of inventing a new fundamental learning algorithm.

Keywords: business analytics; business intelligence; decision support systems; retail analytics; profit prediction; Random Forest; Gradient Boosting; hybrid modelling; explainable AI; feature selection; interactive dashboard

1. INTRODUCTION

1.1 Background

Business intelligence (BI) has evolved from static reporting toward analytical environments that combine data preparation, multidimensional analysis, visualization, predictive modelling, and decision support. Foundational BI research emphasizes the relationship between data, analytical technologies, applications, and organizational decision making [1]. Data warehousing and OLAP research established the importance of structured analytical data and multidimensional querying [2], while later work highlighted the organizational and managerial conditions required for BI systems to create value [3], [4].

In contemporary retail and e-commerce environments, decision makers must interpret sales, profit, discount, product, regional, customer, and operational information simultaneously. A dashboard that only reports historical KPIs may not answer predictive questions, while a predictive model without explanation can be difficult for business users to trust. The present work therefore treats visualization, prediction, explanation, and recommendation as connected stages of one decision-support workflow.

1.2 Problem Statement

Retail transaction datasets contain nonlinear relationships, categorical variables, heterogeneous scales, and business effects such as discounts and product categories. A fragmented workflow can produce separate EDA charts, model metrics, and recommendations without maintaining a clear evidence chain from data to decision. The research problem addressed in this study is: How can an integrated business analytics system combine reproducible data preparation, exploratory analysis, predictive modelling, explainability, hybrid-model evaluation, and interactive visualization so that analytical evidence can be translated into defensible business decisions?

1.3 Research Gap

Existing literature provides mature techniques for BI, predictive modelling, ensemble learning, and explainable AI. Retail studies also demonstrate that ensembles can outperform individual models in forecasting and profitability-related tasks [5]-[9]. However, the reviewed literature uses different datasets, targets, and experimental designs, and a published hybrid that performs well in one retail problem cannot automatically be assumed to be optimal for transaction-level profit prediction. In addition, the current project requires the hybrid to be formed through an explicit sequence of baseline reproduction, limitation identification, alternative hybrid construction, validation, and comparative testing.

The gap addressed in this study is therefore methodological and application-specific: the project constructs a traceable decision-support pipeline in which multiple candidate model combinations are evaluated rather than selecting a hybrid a priori. The paper does not claim that Random Forest, Gradient Boosting, or simple averaging is novel. Instead, novelty is treated as a research question and is bounded by the literature review and the results of the proposed hybrid experiments.

1.4 Objectives

- Develop a reproducible EDA and preprocessing workflow for three heterogeneous business datasets.
- Engineer business-relevant predictors and apply a hybrid feature-selection strategy combining mutual information and Random Forest importance.
- Reproduce and compare multiple baseline regression models for transaction-level profit prediction.
- Study published hybrid and ensemble approaches and use them to motivate multiple project-specific hybrid candidates.
- Evaluate candidate hybrids using validation-based selection and independent test-set metrics.
- Apply explainability techniques to identify influential predictors and connect model evidence to business interpretation.
- Integrate analytics, prediction, explainability, recommendations, and natural-language question routing into an interactive dashboard.

1.5 Research Questions

- RQ1. What data preparation and EDA workflow produces reliable analytical inputs across heterogeneous retail datasets?
- RQ2. Does hybrid feature selection provide a compact and useful predictor set for profit estimation?
- RQ3. How do linear and nonlinear baseline models compare for transaction-level profit prediction?
- RQ4. How do alternative hybrid constructions behave relative to their component models?
- RQ5. Which model inputs are most influential according to model-agnostic explainability?
- RQ6. How can predictive and explanatory evidence be exposed through an interactive business decision-support interface?

2. LITERATURE REVIEW AND MODEL STUDY

This section follows the faculty-prescribed literature-study structure: (i) study existing published models; (ii) identify datasets, baselines, metrics, and hybrid formation mechanisms; (iii) reproduce relevant baseline families; (iv) identify limitations or gaps; (v) construct more than one candidate hybrid; and (vi) compare the candidates before selecting the final model. The literature review is therefore not a generic list of papers; it is used to justify the experimental design.

2.1 Published Studies and Their Experimental Designs

Study	Task	Dataset / Baselines	Hybrid / Ensemble	Metrics	Relevance to this project
Mitra et al. (2022) [5]	Retail sales/demand forecasting	US retail weekly sales; RF, XGBoost, GB, AdaBoost, ANN	RF-XGBoost-LR hybrid	MAE, MSE, RZ	Hybrid combines RF and XGBoost predictions with LR; reported to outperform individual models.
Kovářová et al. / retail profitability study (2023) [6]	Retail firm profitability	551 retail-oriented firms; panel data	Regression, RF, DNN, LSTM, portfolios, ensembles	Predictive error measures, ANOVA	Ensembles/portfolios were reported as more accurate and stable than individual methods.
Lessmann et al. (2021) [7]	Profit-oriented customer targeting	25 real-world marketing datasets	Profit-conscious ensemble selection	Accuracy and campaign profit	Model selection explicitly incorporates statistical and economic objectives.
Winkler et al. (2023) [8]	Retail demand forecasting	Large-scale grocery demand data	Extra Trees vs LSTM/deep learning	Forecasting error metrics	Tree ensembles and deep learning were compared under a real retail forecasting setting.
Samal & Ghosh (2026) [9]	Multi-channel retail demand forecasting	Online/offline FMCG retail time series	Attention-BiLSTM + XGBoost with residual correction	Multiple forecasting errors	Hybrid combines sequential representation with feature-based boosting and uses residual correction.
Lakshmi & Rayalu (2026) [10]	AQI prediction; methodological hybrid evidence	Meteorological/pollution data	RF + Gradient Boosting voting; stacking	RMSE, MAE, MAPE, RMSLE, RZ	Demonstrates that RF+GB averaging and stacking are established hybrid patterns; therefore simple RF+GB averaging is not treated as novel here.

2.2 What the Published Authors Actually Do

- Mitra et al. [5] explicitly compare several individual regression learners before introducing an RF-XGBoost-LR hybrid. This supports the faculty requirement that a proposed combination should be evaluated against its component models and other baselines.
- The retail profitability study [6] compares econometric and machine-learning approaches and reports that portfolio/ensemble strategies can improve predictive stability. This motivates evaluating multiple ensemble constructions rather than assuming a single learner is optimal.
- Lessmann et al. [7] demonstrate that business objectives can be incorporated into model selection rather than optimizing predictive accuracy alone. This is directly relevant to a business decision-support project because model performance must eventually be interpreted in a business context.
- Samal and Ghosh [9] show a recent residual-correction hybrid in retail forecasting. The residual-correction idea motivates the project experiment in which one model predicts the main target and a second model learns the remaining error pattern.
- Lakshmi and Rayalu [10] explicitly use RF and Gradient Boosting in a voting regressor. Consequently, a simple 50:50 RF+GB average cannot be presented as the novel contribution of this project.

2.3 Literature-Based Gap and Proposed Direction

The literature suggests three important design principles. First, strong baseline models must be established before a hybrid is introduced. Second, the combination mechanism itself matters: averaging, stacking, residual correction, and objective-aware selection are different methodologies. Third, the proposed method must be evaluated on the project datasets rather than justified only from results reported elsewhere. Based on these principles, the current project

evaluates simple model combinations and more structured residual-correction and constrained-ensemble candidates. The final proposed hybrid will be selected only if it demonstrates a reproducible advantage over the strongest standalone model and satisfies the novelty check.

3. DATASETS AND BUSINESS CONTEXT

3.1 Dataset 1: Tableau Sample Superstore

The first dataset is the Tableau Sample Superstore transaction dataset. It contains order-level retail information including sales, quantity, discount, category, subcategory, region, state/province, shipping information, and profit. It is the primary dataset for transaction-level profit modelling because profit is explicitly available as a target variable.

3.2 Dataset 2: Global Superstore

Global Superstore provides a multinational retail context with transaction-level sales, quantity, discount, shipping, market, region, category, subcategory, state, and profit information. It is used to test whether the modelling workflow behaves consistently under a broader geographic and market structure.

3.3 Dataset 3: Brazilian E-Commerce Public Dataset (Olist)

The currently available Olist file in the project contains customer identity and geographic fields rather than the complete order/payment tables required for transaction-level profit or RFM analysis. Therefore, the project does not fabricate a profit target for Olist. Instead, the available customer geography is used for customer segmentation. This dataset-specific decision preserves methodological validity.

Table 1. Dataset roles in the proposed framework

Dataset	Primary task	Target / analytical object	Reason
Superstore	Profit regression	Profit	Complete transaction-level sales and profit fields.
Global Superstore	Profit regression	Profit	Independent retail dataset for cross-dataset validation.
Olist customer file	Customer segmentation	Geographic customer representation	Available file contains customer/location fields; profit is not present.

4. METHODOLOGY

4.1 Overall Research Pipeline

The implemented pipeline follows the sequence: raw datasets → data quality assessment → descriptive statistics → missing-value and duplicate analysis → outlier analysis → univariate analysis → bivariate analysis → multivariate analysis → correlation analysis → feature engineering → hybrid feature selection → baseline models → literature-informed hybrid candidates → validation and test evaluation → explainability → business recommendations → interactive dashboard. This follows the faculty roadmap, which emphasizes EDA, preprocessing, feature engineering, feature selection, multiple ML models, explainability, visualization, and comparative analysis [Faculty roadmap].

4.2 Data Preparation and Feature Engineering

- Column names were standardized to consistent machine-readable forms.
- Shipping duration was engineered where date fields permitted it.
- Profit margin and calendar variables such as order year and month were derived for analysis.
- Categorical variables were encoded using one-hot encoding within the modelling pipeline.
- Missing numerical values were handled through median imputation and categorical values through most-frequent imputation where required.
- Profit margin was excluded from predictive feature selection to avoid target leakage because it is directly derived from profit and sales.

4.3 Hybrid Feature Selection

The feature-selection stage combines two complementary signals. Mutual information captures nonlinear dependency between candidate predictors and profit, while Random Forest feature importance captures predictive contribution within a nonlinear ensemble. Features supported by both mechanisms are prioritized. For Superstore, the final compact set used in the validated experiment was sales, discount, subcategory, state/province, category, quantity, region, and order month.

4.4 Baseline Models

- Naive baseline: provides a simple reference point.
- Ridge Regression: regularized linear baseline.
- Random Forest: bagged nonlinear tree ensemble.
- Gradient Boosting: sequential additive tree ensemble.
- Feature-selection-aware Random Forest: tests whether the compact feature set changes predictive behaviour relative to the broader feature representation.

4.5 Hybrid Candidate Design

Following the literature, multiple candidate combinations are considered rather than selecting one hybrid in advance. Completed experiments include RF+GB prediction blending, Ridge+GB, Ridge+RF, a three-model RF+GB+Ridge combination, a validation-weight search, and GB+Ridge residual correction. The next experiment extends this study to structured residual correction and constrained multi-model blending. A candidate is not accepted as the proposed hybrid solely because it combines multiple models; it must be supported by validation and independent test evidence.

4.6 Evaluation Metrics

Regression performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), symmetric MAPE (sMAPE), and coefficient of determination (RZ). MAE gives average absolute error, RMSE penalizes larger errors more strongly, MAPE provides a relative-error view but can become unstable near zero targets, sMAPE reduces some of that instability, and RZ measures explained variance. For multi-model selection, the project also uses average rank across the metrics so that a model is not declared best from one metric alone.

4.7 Explainability

Permutation importance is used as a model-agnostic global explanation. The method measures the change in predictive performance when a feature is randomly permuted. Higher performance degradation indicates greater model dependence. The interpretation is predictive rather than causal; correlated predictors can distribute importance across one another.

4.8 Decision-Support Layer and Dashboard

The Streamlit dashboard exposes dataset selection, synchronized filters, KPI cards, trend and category analysis, profit prediction, explainability, business recommendations, model comparison, and an Ask-AI interface. The natural-language layer is intentionally rule-based: it maps supported business intents to deterministic analytical operations rather than claiming unrestricted conversational intelligence.

5. EXPERIMENTAL RESULTS

5.1 Dataset 1 — Superstore

Model	Family	MAE	RMSE	MAPE	sMAPE	R^2	Rank
Gradient Boosting	Baseline	21.7034	93.7295	104.4600	53.5164	0.8369	1
Hybrid 1 — RF + GB	Hybrid candidate	24.4975	102.2760	128.9554	57.0464	0.8059	2
Hybrid 6 — GB + Ridge residual	Hybrid candidate	27.4167	92.7034	248.5078	85.4185	0.8405	3
RF — selected features	Baseline	30.6196	130.4853	163.8832	61.9141	0.6840	4

Hybrid 5 — validation optimized	Hybrid candidate	30.6196	130.4853	163.8832	61.9141	0.6840	4
Hybrid 4 — RF + GB + Ridge	Hybrid candidate	33.1179	113.7438	288.0668	90.6288	0.7599	6
Hybrid 3 — Ridge + GB	Hybrid candidate	36.8252	114.4703	388.8591	100.6583	0.7568	7
Random Forest	Baseline	38.3207	147.0539	245.4373	78.1313	0.5986	8
Hybrid 2 — Ridge + RF	Hybrid candidate	41.0699	133.5664	400.7525	102.4267	0.6689	9
Ridge Regression	Baseline	57.4713	152.8553	728.6297	120.6365	0.5663	10
Naive	Baseline	65.4980	232.1307	577.0504	123.2188	-0.0001	11

The Superstore results show that Gradient Boosting is currently the strongest completed model by average rank. Hybrid 1 (RF+GB) is the strongest genuine hybrid among the completed candidates by average rank, but it does not outperform Gradient Boosting. Hybrid 6 obtains a slightly better RMSE and RZ than Gradient Boosting, but its MAPE and sMAPE are substantially worse. This is precisely why the final hybrid cannot be selected from a single metric.

5.2 Dataset 2 — Global Superstore

Model	Family	MAE	RMSE	MAPE	sMAPE	R^2	Rank
Gradient Boosting	Baseline	35.5769	94.1239	552.6496	69.7579	0.6671	1
Hybrid 5 — validation optimized	Hybrid candidate	35.5769	94.1239	552.6496	69.7579	0.6671	1
Hybrid 1 — RF + GB	Hybrid candidate	37.1724	99.8001	595.0656	71.7622	0.6257	3
Hybrid 6 — GB + Ridge residual	Hybrid candidate	38.0971	94.2085	741.0949	82.3451	0.6665	4
RF — selected features	Baseline	40.9543	110.3870	—	—	0.5421	5
Hybrid 4 — RF + GB + Ridge	Hybrid candidate	42.4477	108.9696	—	—	0.5537	6
Random Forest — all	Baseline	43.2267	116.4430	—	—	0.4904	7
Hybrid 3 — Ridge + GB	Hybrid candidate	45.0361	110.6843	—	—	0.5396	8
Hybrid 2 — Ridge + RF	Hybrid candidate	47.5882	119.7187	—	—	0.4614	9
Ridge Regression	Baseline	58.7163	135.1317	—	—	0.3137	10
Naive	Baseline	66.4684	163.1214	—	—	~0	11

Global Superstore again places Gradient Boosting at the top of the completed comparison. The validation-optimized ensemble collapsed to a 100% Gradient Boosting weight, which is an important negative result: the optimization procedure did not justify a genuine hybrid on this dataset. Hybrid 1 and Hybrid 6 remain competitive but do not establish a consistent superiority over Gradient Boosting.

5.3 Dataset 3 — Customer Geographic Segmentation

Because the available Olist customer file does not contain transaction profit, it is treated as an unsupervised segmentation problem. K-Means, Agglomerative Clustering, Gaussian Mixture Models, and DBSCAN were evaluated. The best completed configuration was DBSCAN with eps=0.30 and min_samples=5, producing 27 clusters with a silhouette score of 0.9143, Davies–Bouldin index of 0.0820, and Calinski–Harabasz score of 111738.37 on the deterministic evaluation sample. The high silhouette and low Davies–Bouldin value indicate strong geometric separation under the engineered geographic representation.

6. COMPARISON WITH PUBLISHED MODELING APPROACHES

The project results are interpreted against the published literature rather than compared numerically across incompatible datasets. Mitra et al. [5] reported an RF-XGBoost-LR hybrid for retail sales forecasting; the current project does not reproduce that exact architecture and therefore does not claim a direct numerical comparison. The 2023 retail profitability study [6] supports the use of RF and ensemble methods for profitability prediction. Lessmann et al. [7] demonstrate that business objectives can be integrated into ensemble selection. Samal and Ghosh [9] provide

recent evidence for residual-correction hybridization in retail forecasting. Lakshmi and Rayalu [10] explicitly establish RF+GB voting as a published hybrid pattern, which prevents this project from claiming simple RF+GB averaging as novel.

Table 2. Literature-to-project mapping

Literature pattern	Status in project	Implication
Random Forest baseline	Implemented	Strong nonlinear baseline and explainability source.
Gradient Boosting baseline	Implemented	Current strongest completed standalone model on D1 and D2.
RF + GB voting/averaging	Implemented as candidate	Useful benchmark, but not novel because the pattern is published [10].
RF-XGBoost-LR stacking	Reviewed in literature; not yet reproduced	Should be implemented if included as a final literature benchmark.
Residual correction	Implemented in candidate form; extended experiment pending	Promising research direction because it modifies the prediction pathway rather than simply averaging outputs.
Business-objective-aware model selection	Conceptually incorporated in decision-support layer	Requires a formal business utility objective for a stronger future experiment.

7. EXPLAINABLE AI AND BUSINESS INSIGHTS

7.1 Feature Importance

Permutation importance on the trained Superstore Random Forest identified sales and discount as the dominant predictors, followed by subcategory and state/province. The measured importance values were: sales 1.607752, discount 1.064891, subcategory 0.056778, state/province 0.017471, category 0.011978, quantity 0.010081, region 0.006637, and order month 0.006349. The large importance of sales and discount indicates that the fitted model relies heavily on transaction value and pricing reduction when estimating profit.

7.2 Business Findings

- Total sales: 2,326,534.35.
- Total profit: 292,296.81.
- Total quantity: 38,654 units.
- Average discount: approximately 0.1554.
- Average profit margin: approximately 12.22%.
- Technology generated the highest category-level profit in the completed business analysis.
- Furniture generated the lowest category-level profit among the main categories.
- West was the highest-profit region in the completed analysis.
- Tables, Bookcases, and Supplies were identified as loss-making subcategories.
- The 40–60% discount range had the lowest average profit per transaction in the completed analysis.

These observations are descriptive associations from the project data. They should not be interpreted as causal estimates without a controlled causal design.

8. INTEGRATED DECISION-SUPPORT SYSTEM

The dashboard is designed as the operational layer of the research artifact. The Overview page presents KPIs and high-level trends. Analytics provides category, region, subcategory, discount, and temporal views. Profit Prediction exposes the trained regression model where a transaction-level profit target is available. XAI presents model-based feature importance. Recommendations translate descriptive patterns into deterministic business suggestions. The Ask-AI layer maps supported natural-language questions to predefined analytical functions. The ML Models section provides comparative model metrics. Dataset 3 uses clustering rather than profit regression because of its available schema.

8.1 Decision Traceability

A central design principle is traceability: every recommendation should be linked to an observable metric, model output, or analytical pattern. For example, a recommendation concerning high discounts can be traced to discount distribution, average profit by discount band, and the model's high permutation importance for discount. This is preferable to generating unsupported business advice.

9. DISCUSSION

The completed experiments produce an important methodological finding: hybridization is not automatically beneficial. On both Superstore datasets, Gradient Boosting currently outperforms the completed genuine hybrid candidates when the full metric profile is considered. This suggests that the component models are not necessarily complementary enough under the current feature representation, or that simple blending does not exploit their differences effectively. The result is consistent with the faculty requirement that several possible combinations be evaluated before selecting a proposed hybrid.

The literature further explains why a stronger hybrid experiment is required. Published retail work has used RF-XGBoost-LR stacking [5], ensemble/portfolio approaches for profitability [6], profit-conscious ensemble selection [7], and residual-correction mechanisms [9]. Therefore, the project's next hybrid should be justified as a specific modification to the prediction pathway and should be evaluated against the strongest completed baseline, not merely described as a combination of familiar algorithms.

9.1 Current Research Contribution Boundary

- The project does not claim to invent Random Forest, Gradient Boosting, Ridge Regression, permutation importance, or simple voting.
- The project does not claim RF+GB averaging as a novel hybrid because this combination is already documented in published hybrid ensemble work [10].
- The defensible project contribution is the integrated evidence-traceable pipeline plus the systematic evaluation of multiple hybrid candidates on the selected business datasets.
- A stronger algorithmic novelty claim should only be made after the final hybrid experiment and a broader novelty search.

10. LIMITATIONS AND VALIDITY

- The current regression evidence is based on deterministic train/validation/test splits rather than repeated cross-validation or time-aware validation.
- MAPE can be unstable when actual profit values are close to zero; therefore MAE, RMSE, sMAPE, and RZ are considered jointly.
- The datasets are not merged because their schemas and analytical objects differ; this limits cross-dataset statistical pooling but improves semantic validity.
- The available Olist customer file supports geographic segmentation but not transaction-level profit modelling.
- Permutation importance describes model dependence and does not establish causality.
- The natural-language interface is deterministic and intent-routed rather than a general-purpose conversational AI system.
- No controlled user study has yet been conducted to establish improvements in human decision quality.
- The final proposed hybrid must not be declared until the pending final hybrid experiment is executed and the literature novelty check is completed.

11. CONCLUSION

This paper presents an integrated AI-assisted data visualization and business decision-support framework that connects EDA, feature engineering, hybrid feature selection, predictive modelling, explainable AI, business recommendations, and interactive visualization. The work follows a literature-informed experimental process rather

than treating the dashboard as an isolated visualization artifact. Superstore and Global Superstore provide transaction-level profit modelling, while the available Olist customer data is used for geographic segmentation.

The current validated results demonstrate that Gradient Boosting is the strongest completed standalone model on both regression datasets. Superstore achieved MAE 21.7034, RMSE 93.7295, sMAPE 53.5164, and RZ 0.8369, while Global Superstore achieved MAE 35.5769, RMSE 94.1239, sMAPE 69.7579, and RZ 0.6671. Multiple hybrid candidates have been evaluated, but none of the completed genuine hybrids has yet established consistent superiority over Gradient Boosting. This negative result is scientifically important because it prevents an unsupported claim of hybrid superiority.

The final stage of the research is therefore to execute the structured residual-correction and constrained-ensemble experiment, compare it against the strongest baselines and literature-informed candidates, and select the proposed hybrid only if the evidence supports it. The final manuscript should then replace the explicitly marked pending statement with the measured final hybrid results, statistical/visual comparison, and verified novelty statement.

DECLARATIONS

Funding

No external funding was reported for this project. If institutional or sponsored funding applies, the relevant funding statement should be inserted after author confirmation.

Conflict of Interest

The authors declare no conflict of interest, subject to confirmation by all listed authors.

Ethics Statement

The project uses publicly available retail and e-commerce datasets and does not intentionally process personally identifiable human-subject research data. Any institutional ethics requirement should be confirmed according to the final submission rules and data sources.

Data Availability

The final paper should provide the verified repository location for the project code and the legally shareable dataset sources. Where redistribution is restricted, the paper should link to the original dataset provider rather than redistribute the raw data.

Author Contributions

Lellapalli Suvarchala Mrudani: project implementation, data preparation, machine-learning experimentation, dashboard integration, analysis, and manuscript preparation. Additional author contributions: PydahHasitha Sai Keerthana, SeeramSai Sri Kruthi, and K Swapnika. The final contribution statement should be confirmed by all authors before submission.

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12. REPRODUCIBILITY AND FACULTY-SUBMISSION CHECKLIST

Faculty requirement	Current status
Published-model study completed	Yes — literature section identifies datasets, baselines, hybrid mechanisms, and metrics.
Baseline reproduction	Yes — Naive, Ridge, Random Forest, Gradient Boosting and feature-selection-aware RF were evaluated.
Multiple hybrid candidates	Yes — multiple combinations and residual correction candidates were tested.
4–5 model comparison	Yes — the comparative experiments contain substantially more than five

	models/candidates.
Hybrid superiority established	Not yet — current results do not support this claim.
Final hybrid experiment	Pending — structured residual correction and constrained ensemble experiment.
Explainability	Yes — permutation importance and model-specific feature importance are implemented.
Dashboard execution	Implemented — Streamlit dashboard with dataset-specific analytical workflows.
Business recommendations	Implemented — descriptive evidence is converted to deterministic recommendations.
Literature novelty check	Partially completed — simple RF+GB averaging identified as already published; final candidate requires final check.
Plagiarism check	Must be completed before manuscript submission, as required by faculty.
GitHub repository	Must be updated with final code, outputs, requirements, and README before submission.

13. RESULTS: ADDITIONAL TABULAR AND GRAPHICAL OUTPUTS

This section consolidates the principal saved outputs produced by the implementation. It supplements Section 5 with tabular and graphical evidence from exploratory analysis, model comparison, clustering, explainability, and business summaries. Values are reported from the saved project output files; results from different experiment splits are not pooled or treated as directly interchangeable.

13.1 Comparative Regression Results — Dataset 1

Model	MAE ↓	RMSE ↓	MAPE	sMAPE ↓	R^2 ↑	Avg. rank
Gradient Boosting	21.703	93.729	104.460	53.516	0.8369	1.40
Hybrid 1 - RF + GB	24.497	102.276	128.955	57.046	0.8059	2.40
Hybrid 6 - GB + Ridge Residual Correction	27.417	92.703	248.508	85.419	0.8405	3.40
Random Forest - Selected Features	30.620	130.485	163.883	61.914	0.6840	4.90
Hybrid 5 - Validation Optimized Ensemble	30.620	130.485	163.883	61.914	0.6840	4.90
Hybrid 4 - RF + GB + Ridge	33.118	113.744	288.067	90.629	0.7599	5.60
Hybrid 3 - Ridge + GB	36.825	114.470	388.859	100.658	0.7568	6.60
Random Forest	38.321	147.054	245.437	78.131	0.5986	7.20
Hybrid 2 - Ridge + RF	41.070	133.566	400.753	102.427	0.6689	8.60
Ridge Regression	57.471	152.855	728.630	120.636	0.5663	10.20
Naive Model	65.498	232.131	577.050	123.219	-0.0001	10.80

Gradient Boosting ranks first in the completed D1 comparison. RF+GB is the strongest genuine hybrid candidate by average rank among the listed candidates, but it does not outperform Gradient Boosting across the full metric profile.

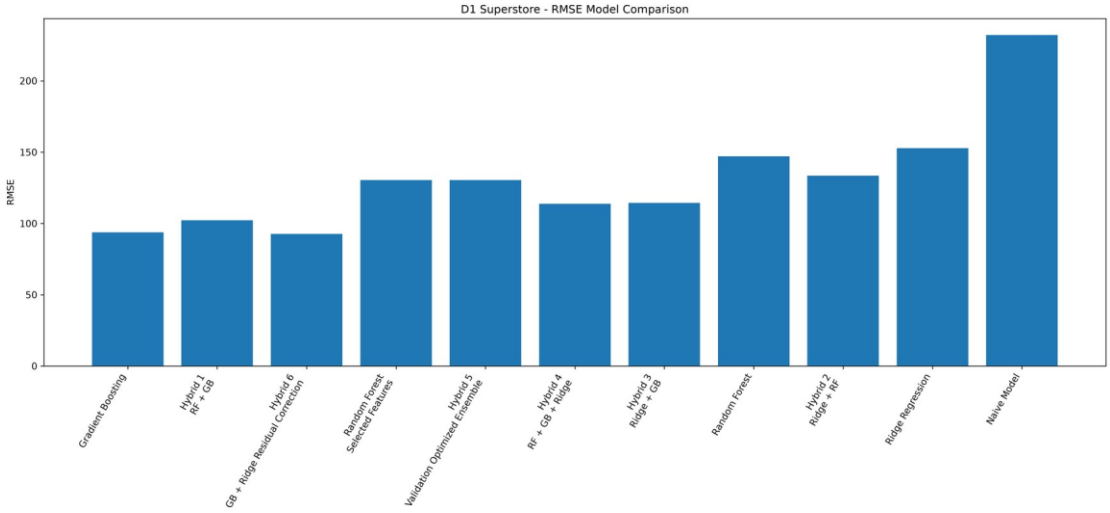


Figure 13.1. Dataset 1 comparative model metrics.

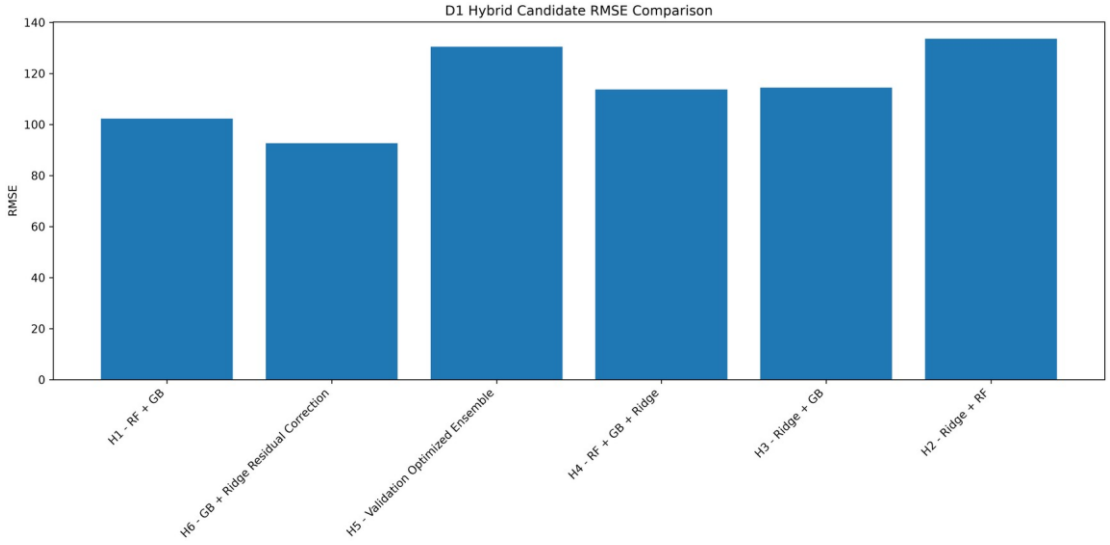


Figure 13.2. Dataset 1 hybrid-candidate comparison.

13.2 Comparative Regression Results — Dataset 2

Model	MAE ↓	RMSE ↓	MAPE	sMAPE ↓	R^2 ↑	Avg. rank
Gradient Boosting	35.577	94.124	552.650	69.758	0.6670	1.50
Hybrid 5 - Validation Optimized Ensemble	35.577	94.124	552.650	69.758	0.6670	1.50
Hybrid 1 - RF + GB	37.172	99.800	595.066	71.762	0.6257	3.40
Hybrid 6 - GB + Ridge Residual Correction	38.097	94.208	741.095	82.345	0.6665	4.20
Random Forest - Selected Features	40.954	110.387	647.945	76.327	0.5421	5.00
Hybrid 4 - RF + GB + Ridge	42.448	108.970	839.538	87.117	0.5537	6.00
Random Forest	43.227	116.443	777.275	80.722	0.4904	6.80
Hybrid 3 - Ridge + GB	45.036	110.684	954.697	92.930	0.5396	7.80
Hybrid 2 - Ridge + RF	47.588	119.719	996.167	94.599	0.4614	9.20
Ridge Regression	58.716	135.132	1390.861	108.076	0.3137	10.20
Naive Model	66.468	163.121	866.857	124.480	-0.0000	10.40

Gradient Boosting is the strongest completed D2 baseline. The validation-optimized ensemble selected the Gradient Boosting model alone, so that result is not evidence of a genuine hybrid advantage.

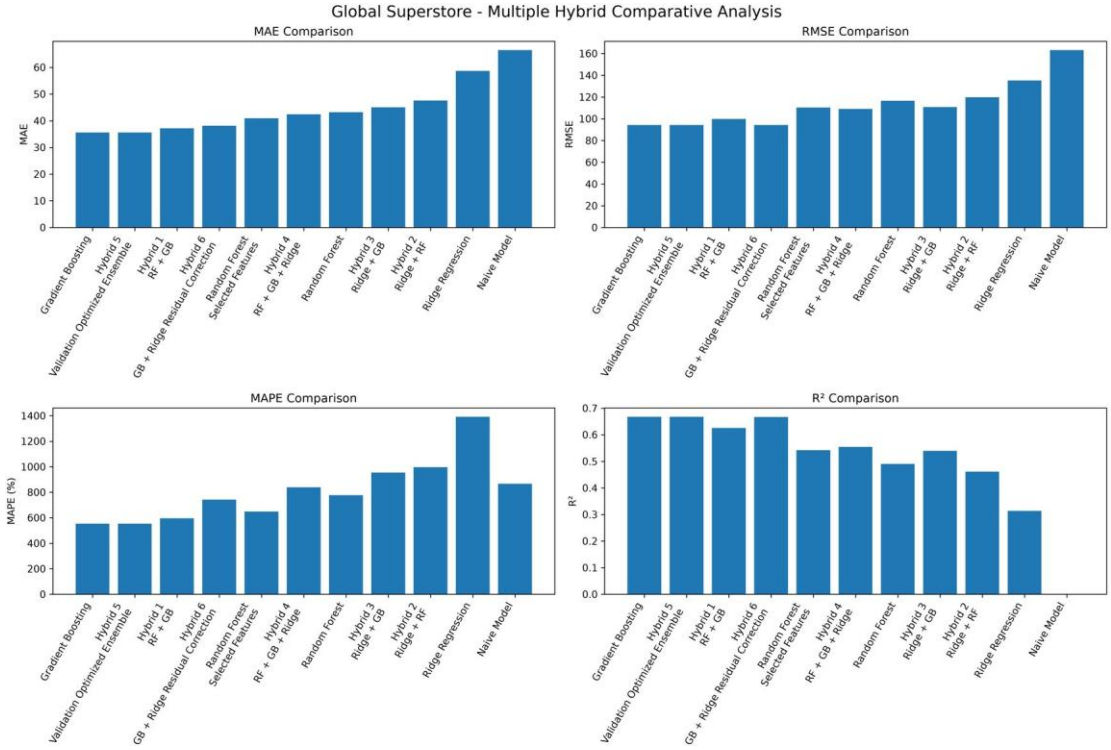


Figure 13.3. Dataset 2 comparative model metrics.

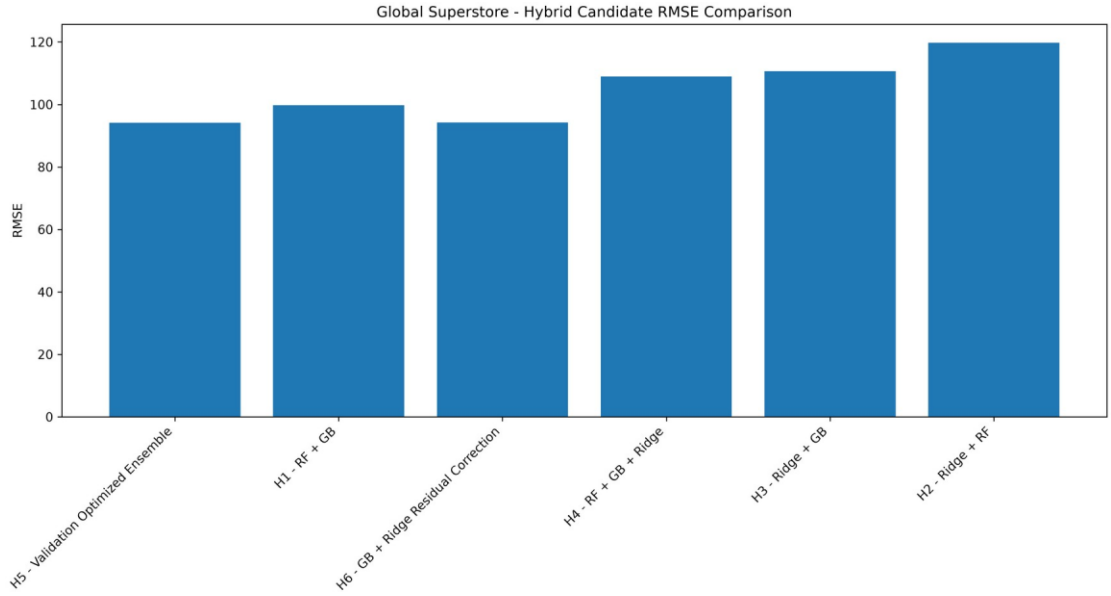


Figure 13.4. Dataset 2 hybrid-candidate comparison.

13.3 Additional Residual-Hybrid Experiment — Dataset 1

Model	MAE	RMSE	sMAPE	R²
Hybrid 1 - GB + RF Residual	15.646	102.800	39.235	0.8352
Random Forest	17.660	131.849	38.780	0.7289
Hybrid 2 - RF + GB Residual	17.660	131.849	38.780	0.7289
Gradient Boosting	23.436	106.964	72.913	0.8216
Hybrid 4 - Constrained GB+RF+Ridge	21.271	125.325	65.537	0.7551
Hybrid 3 - GB + Ridge Residual	24.139	107.463	77.470	0.8199
Ridge Regression	59.160	234.396	118.598	0.1433

This additional experiment used a separate split setup. Its scores should be interpreted within that experiment and not directly ranked against the earlier D1 table unless the split and evaluation protocol are made identical.

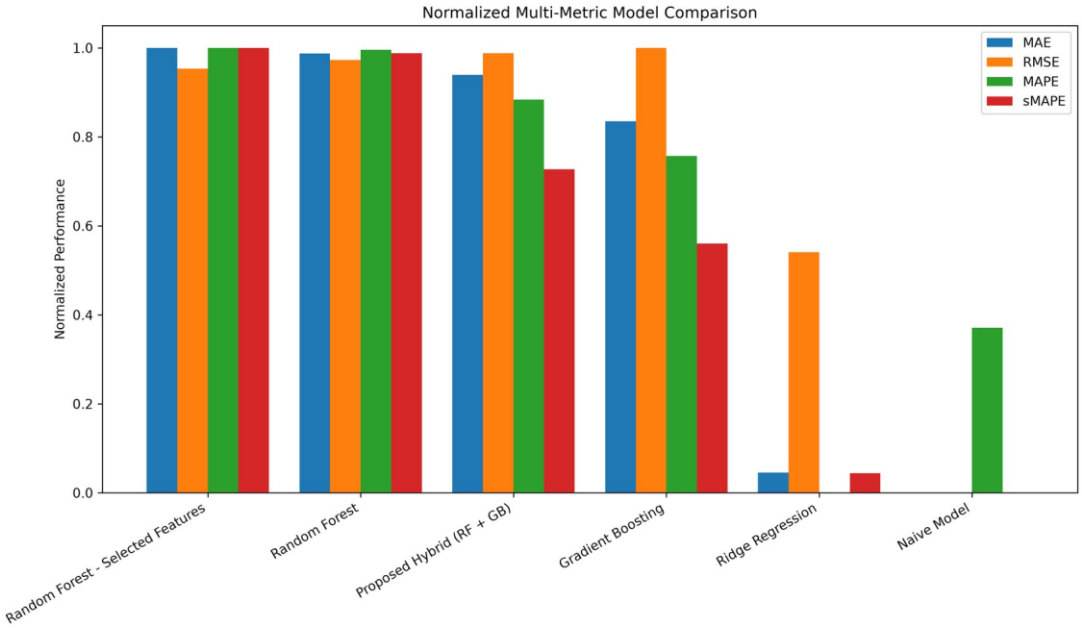


Figure 13.5. Normalized metric comparison from the saved comparative analysis.

13.4 Dataset 3 — Geographic Clustering Results

Method/configuration	Clusters	Noise	Silhouette ↑	Davies–Bouldin ↓	Calinski–Harabasz ↑
DBSCAN (eps=0.30, min_samples=5)	27	0	0.9143	0.0820	111738.37
Agglomerative (k=8)	8	0	0.7714	0.8786	32228.39
K-Means (k=8)	8	0	0.7579	0.9323	30377.52
Gaussian Mixture (k=8)	8	0	0.7367	1.2327	27291.74

DBSCAN (eps=0.30, min_samples=5) achieved the strongest internal scores in the saved evaluation. Because the representation uses ZIP-prefix and state-derived features, this result indicates separation in the engineered feature space and should not be interpreted as proof of geographic distance or business-value segmentation.

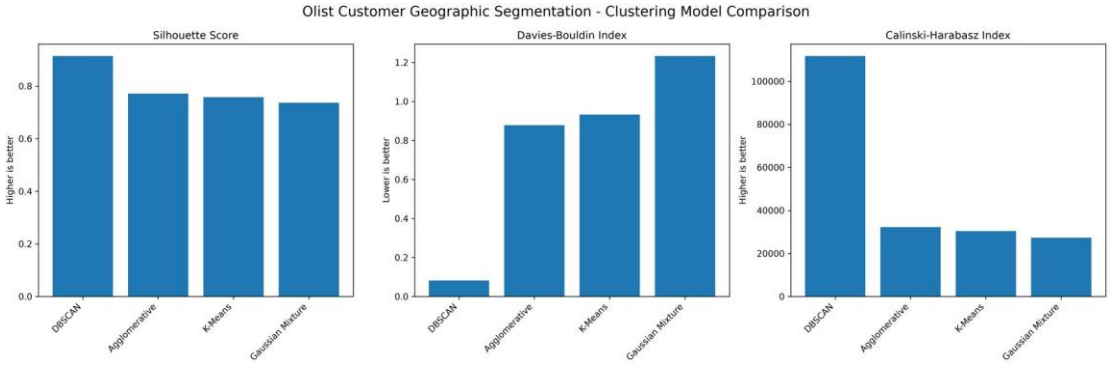


Figure 13.6. Comparison of clustering methods.

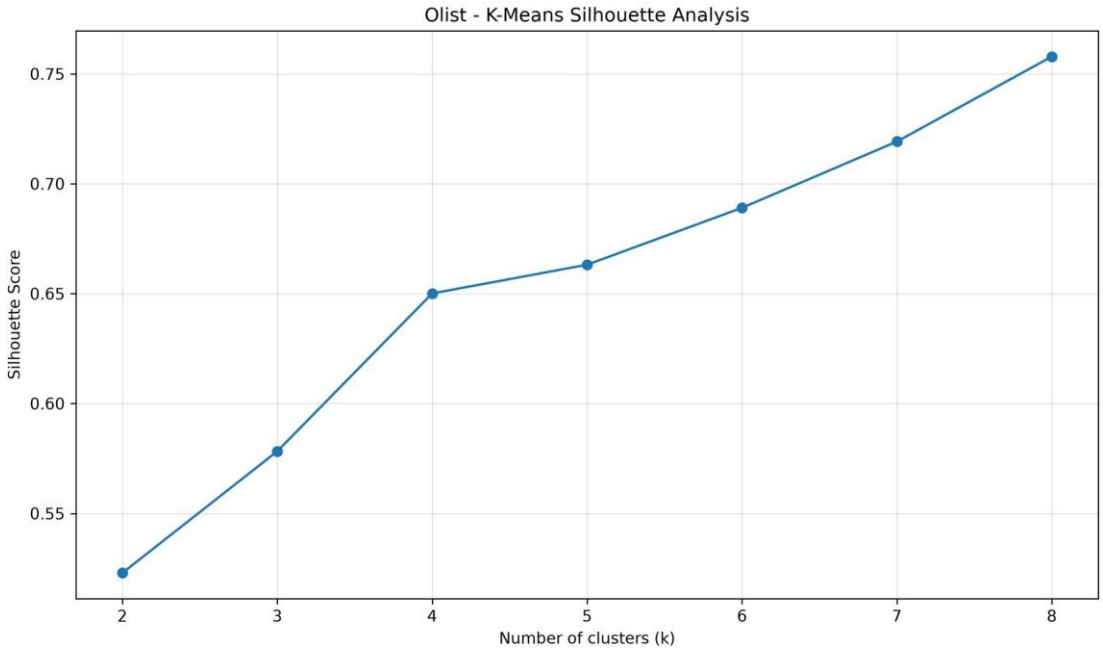


Figure 13.7. K-Means silhouette scores across candidate cluster counts.

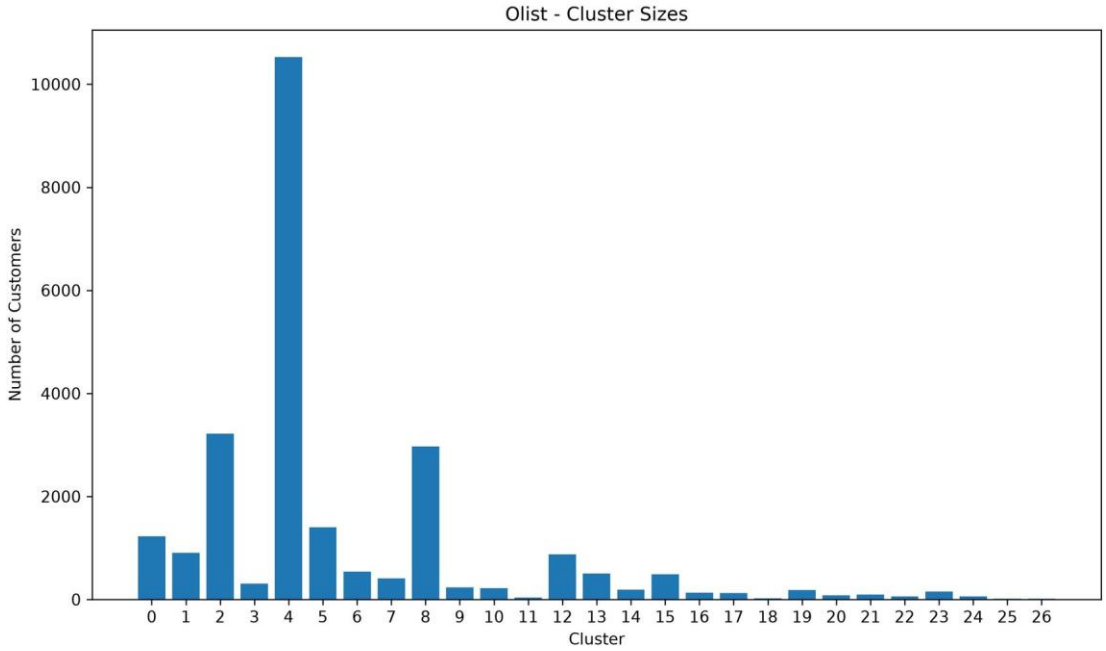


Figure 13.8. Cluster-size output for the selected clustering analysis.

13.5 Feature Selection and Explainability Outputs

Rank	Feature	Mutual information	RF importance	Combined score
1	sales	2.053	0.616	1.000
2	discount	0.482	0.207	0.285
3	subcategory	0.776	0.056	0.234
4	stateprovince	0.292	0.035	0.099
5	category	0.361	0.012	0.097
6	quantity	0.356	0.010	0.095
7	region	0.078	0.016	0.032
8	order_month	0.030	0.015	0.020
9	segment	0.028	0.008	0.014
10	shipping_days	0.003	0.012	0.011

Rank	Feature	Mean importance	Std. deviation
1	sales	1.607752	0.055588
2	discount	1.064891	0.066204
3	subcategory	0.056778	0.001823
4	stateprovince	0.017471	0.001935
5	category	0.011978	0.001154
6	quantity	0.010081	0.001179
7	region	0.006637	0.001473
8	order_month	0.006349	0.001687

Sales and discount have the largest saved permutation-importance values. These values describe model reliance under the evaluation setup and do not establish causal effects.

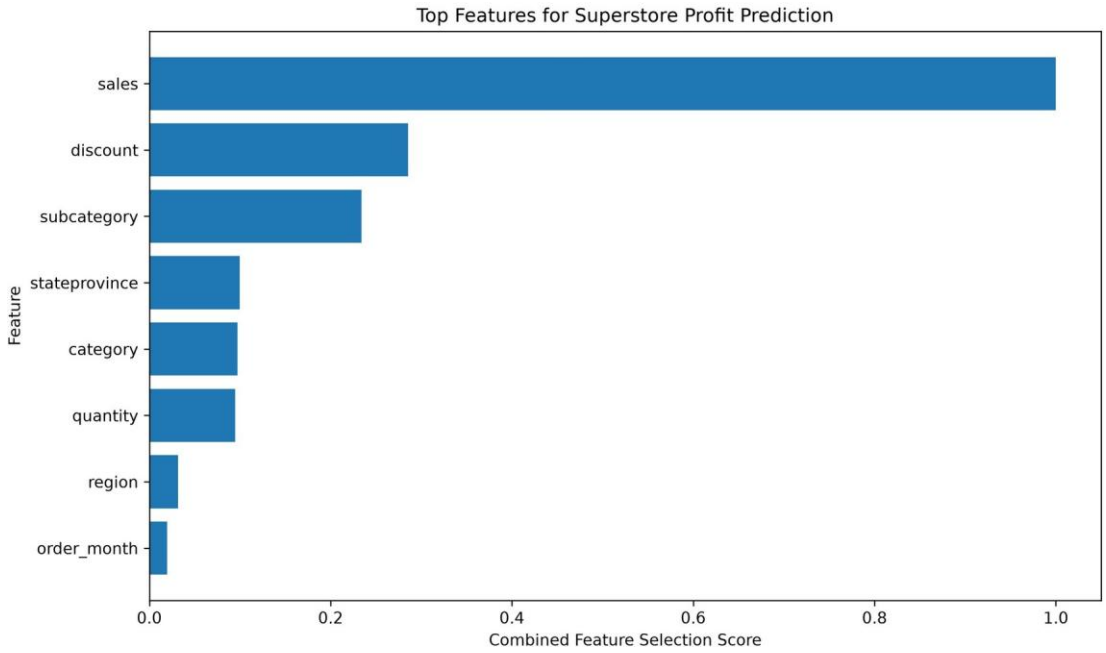


Figure 13.9. Hybrid feature-selection output.

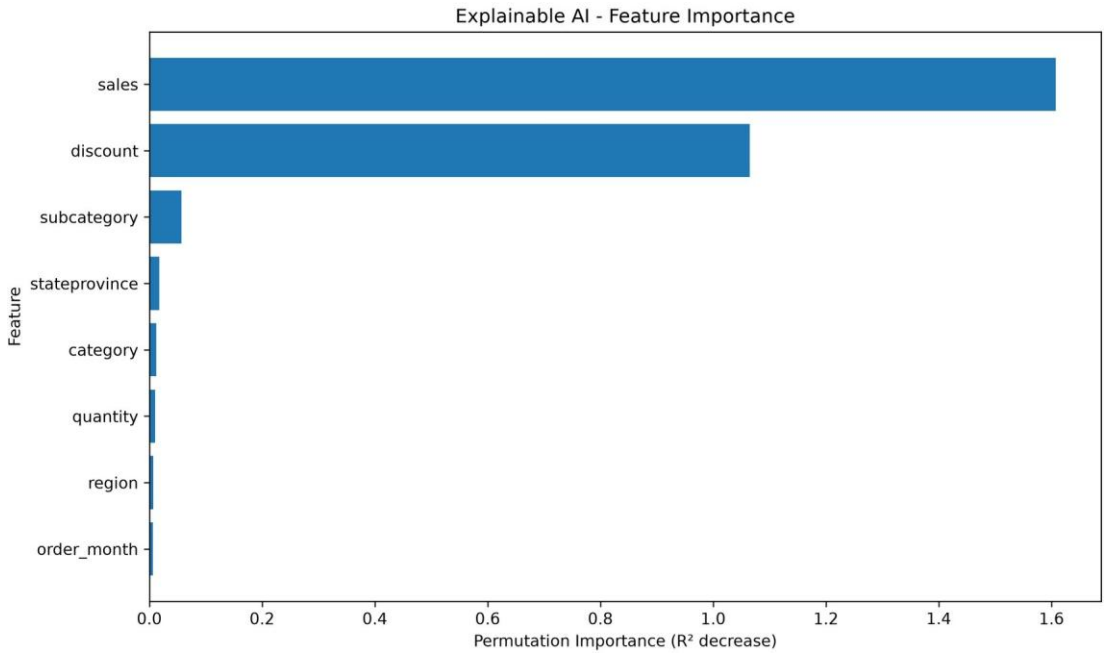


Figure 13.10. Permutation feature-importance output.

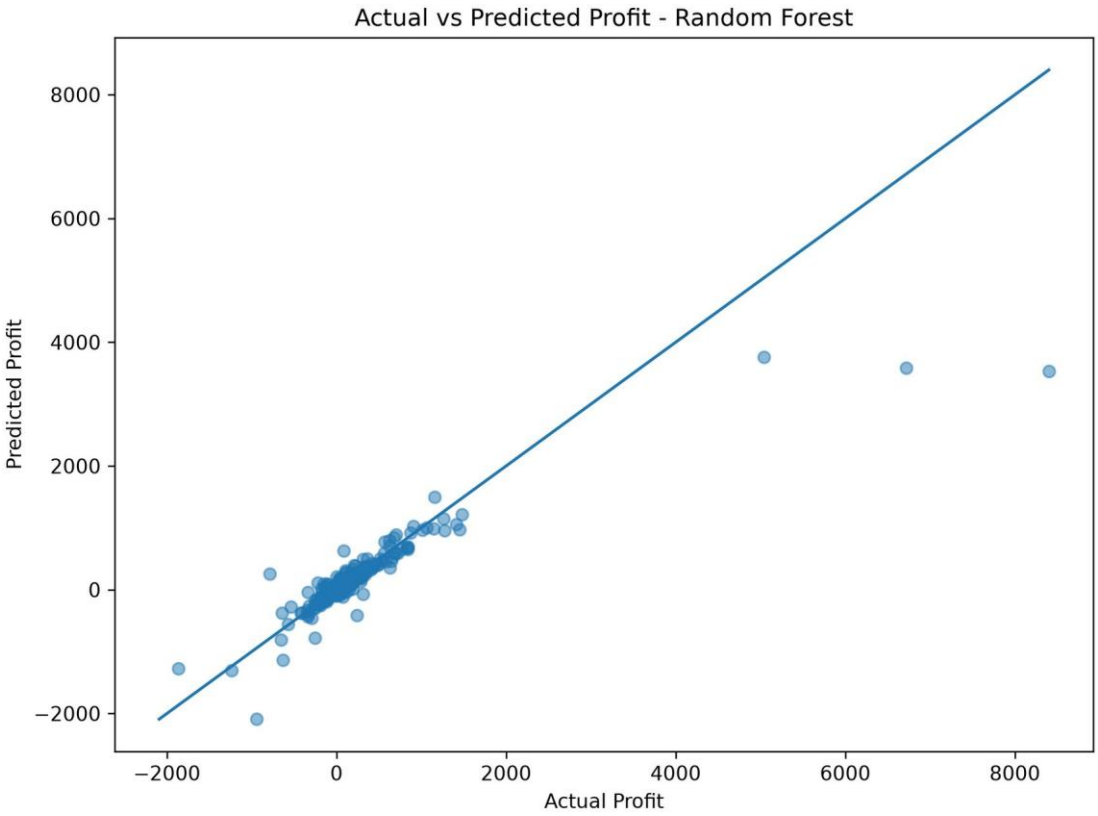


Figure 13.11. Actual versus predicted profit for the saved model.

13.6 Exploratory Data Analysis Outputs

The following figures document the descriptive analysis used to understand the datasets before modelling.

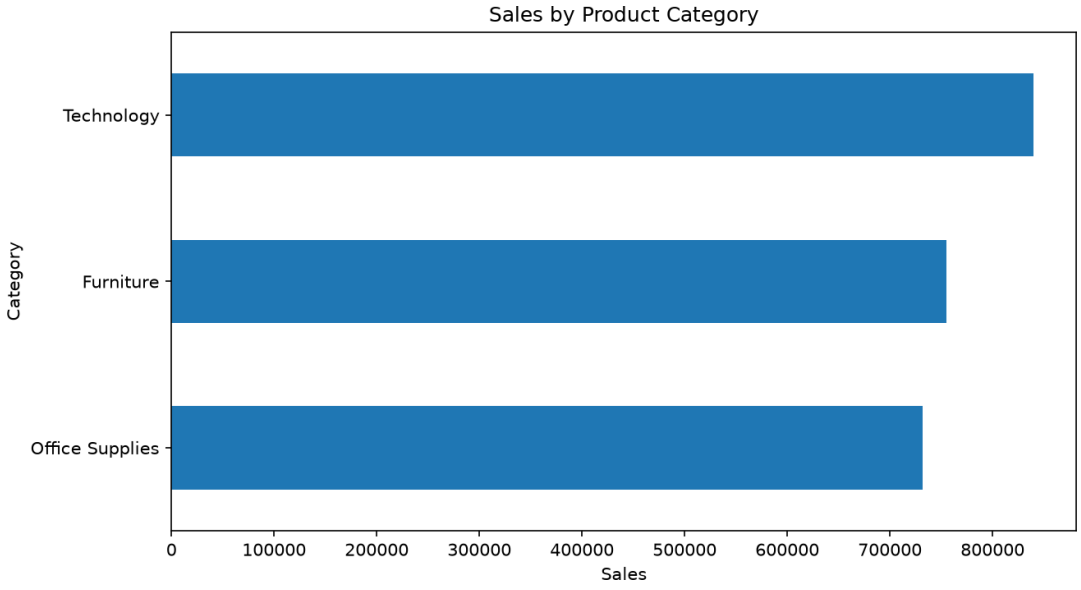


Figure 13.12. Dataset 1 sales by category.

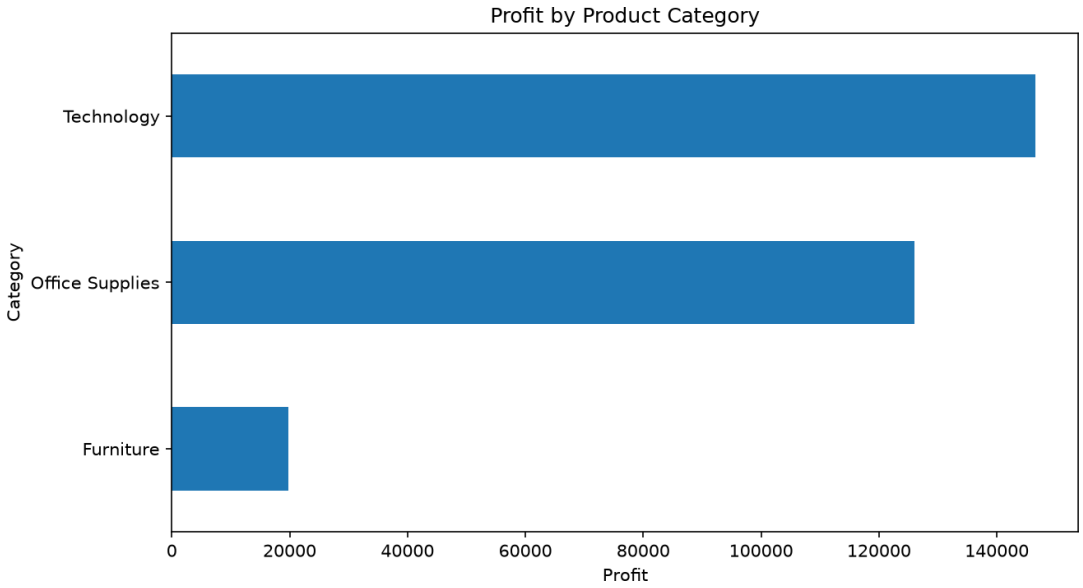


Figure 13.13. Dataset 1 profit by category.

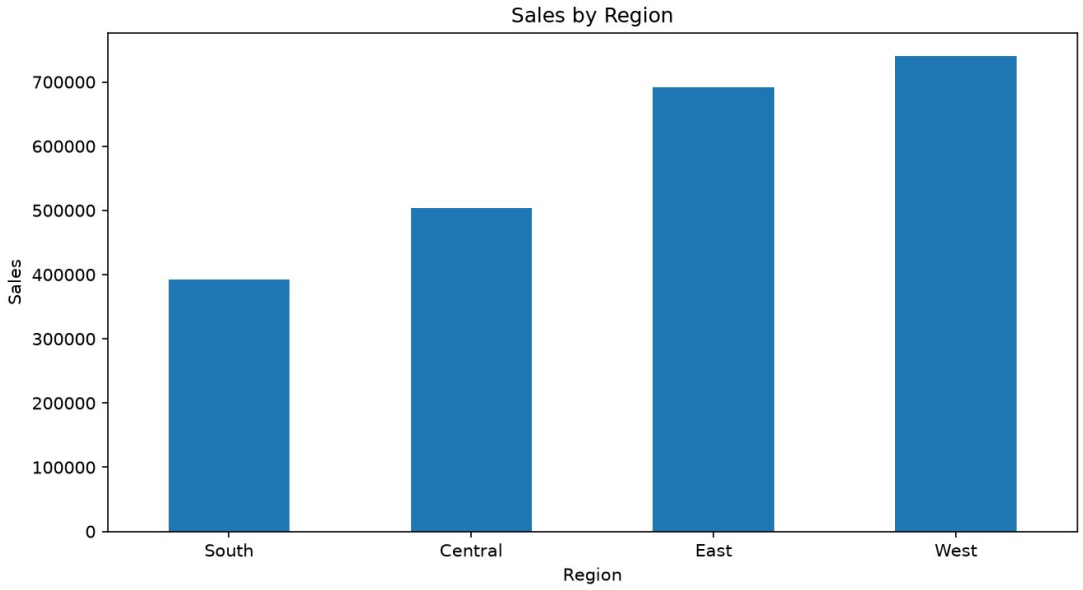


Figure 13.14. Dataset 1 sales by region.

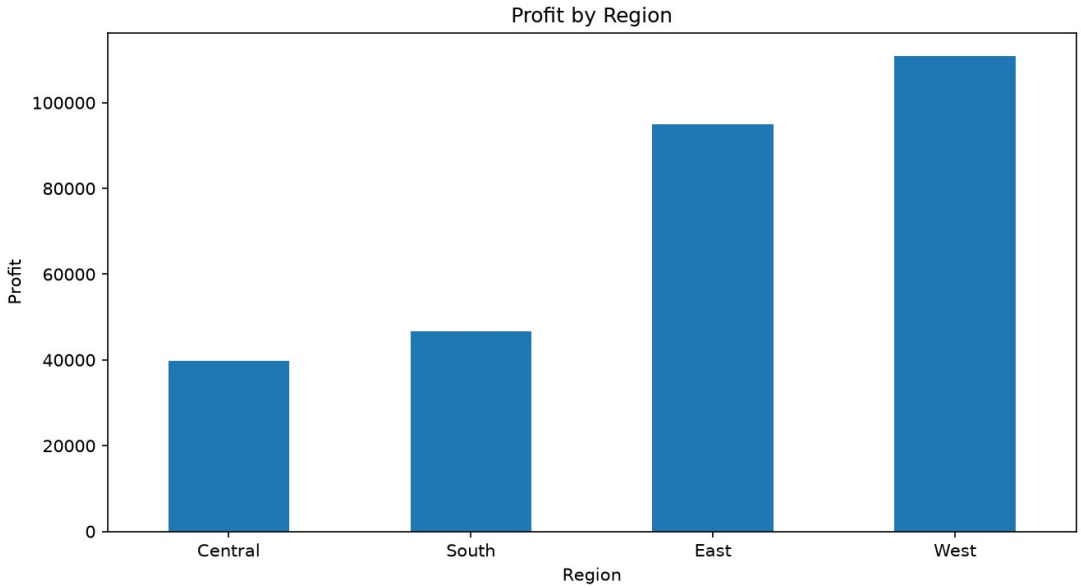


Figure 13.15. Dataset 1 profit by region.

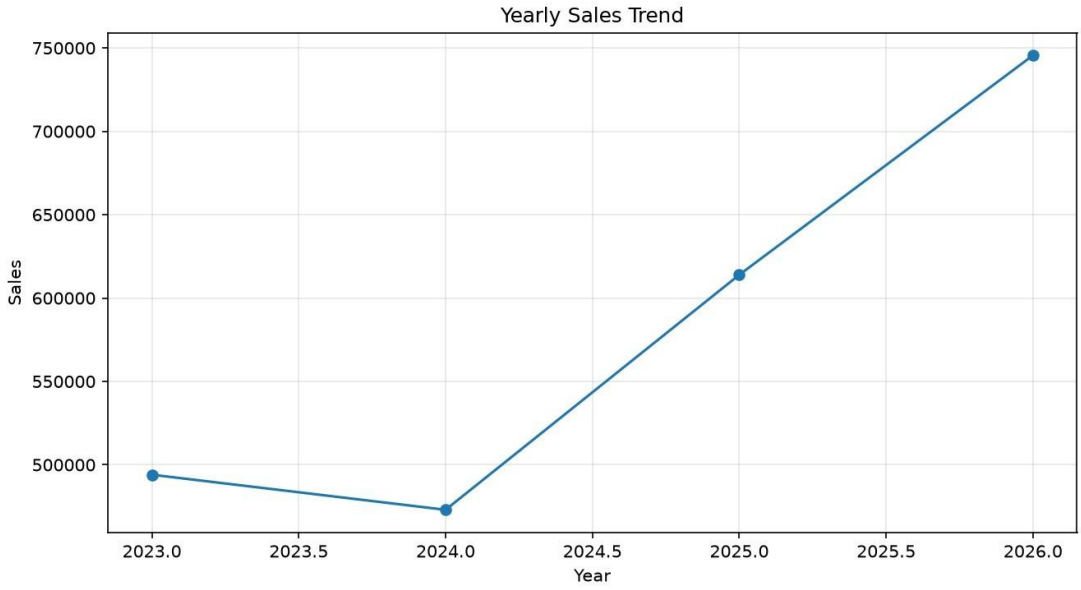


Figure 13.16. Dataset 1 yearly sales trend.

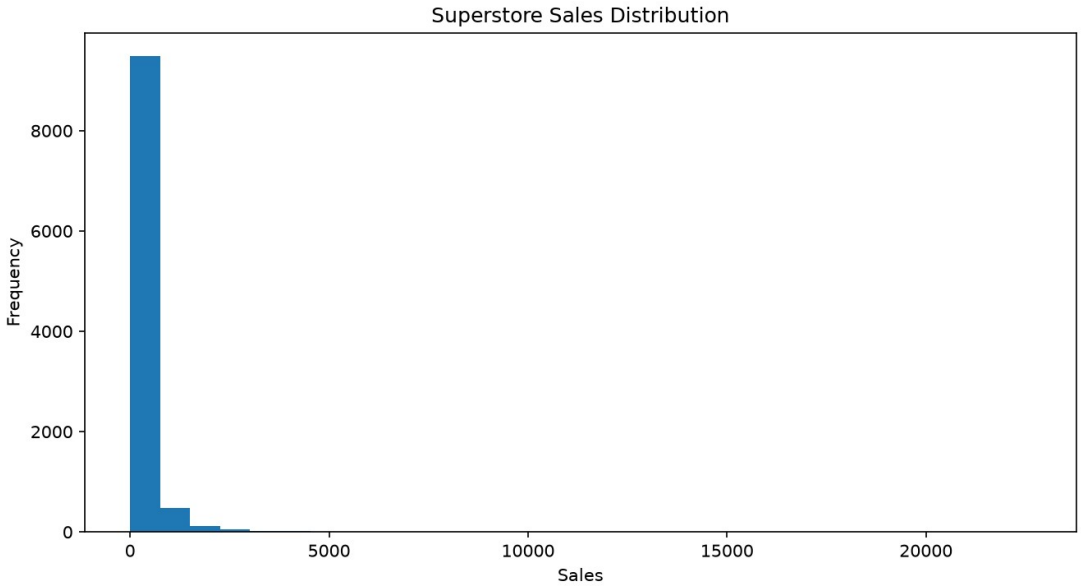


Figure 13.17. Dataset 1 sales distribution.

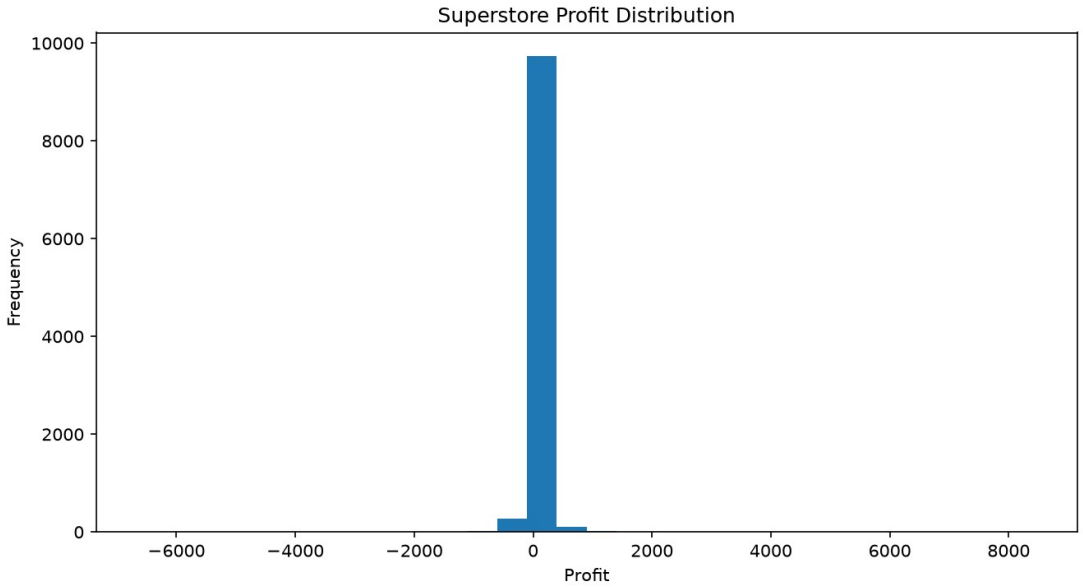


Figure 13.18. Dataset 1 profit distribution.



Figure 13.19. Dataset 1 sales versus profit.

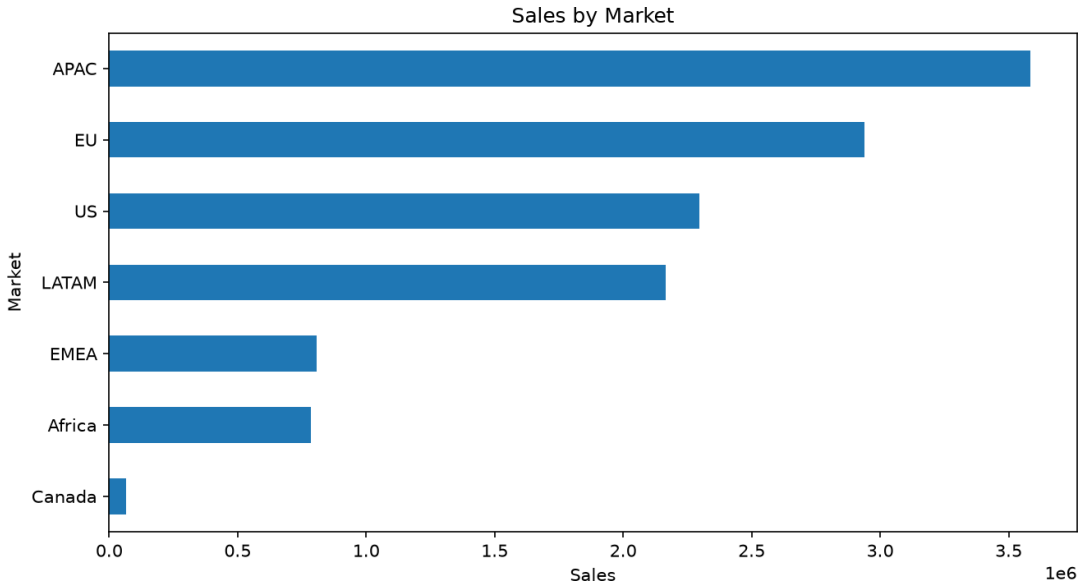


Figure 13.20. Dataset 2 sales by market.

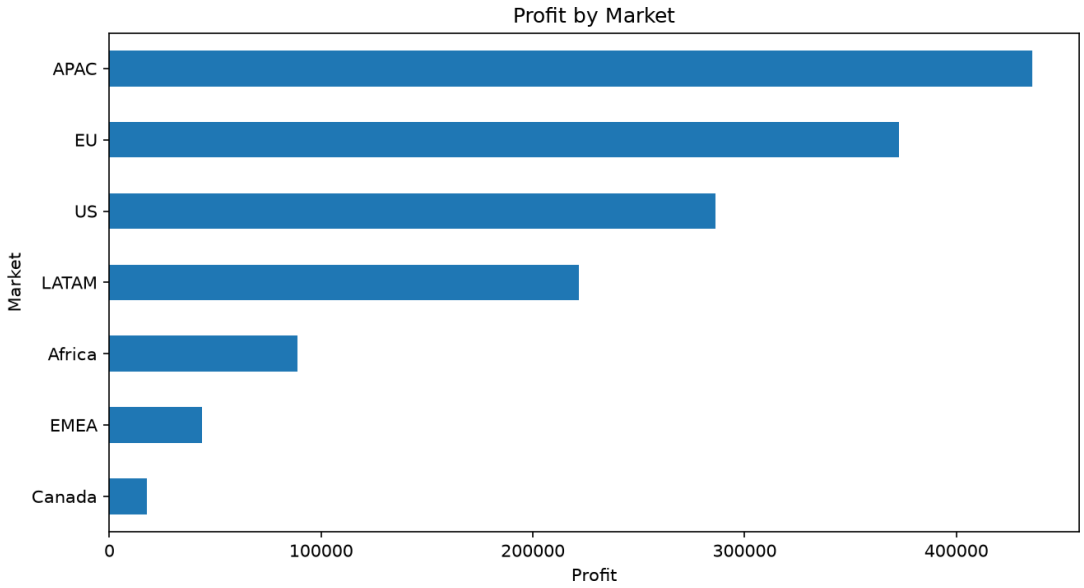


Figure 13.21. Dataset 2 profit by market.

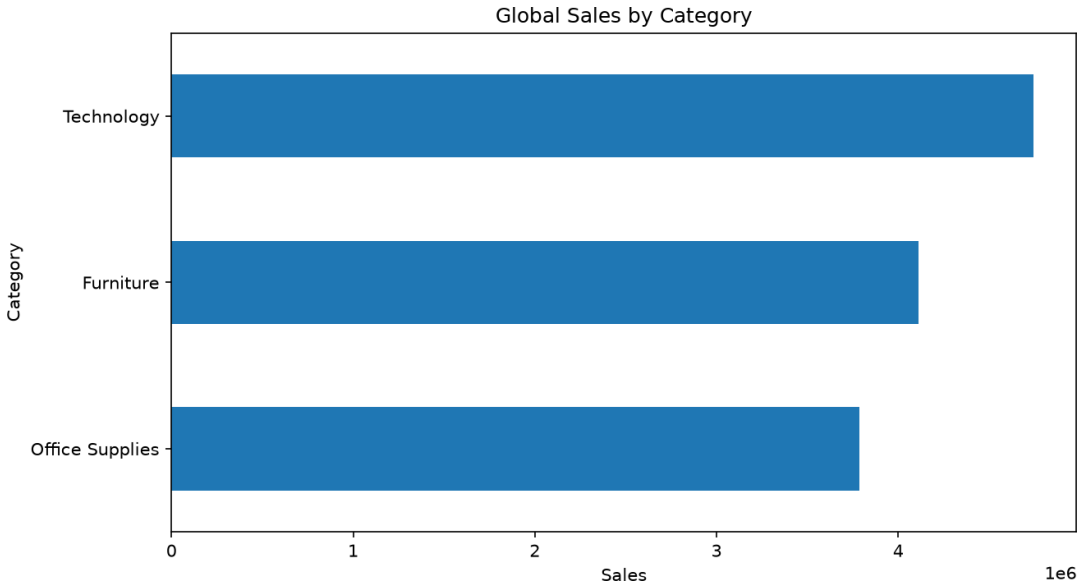


Figure 13.22. Dataset 2 sales by category.

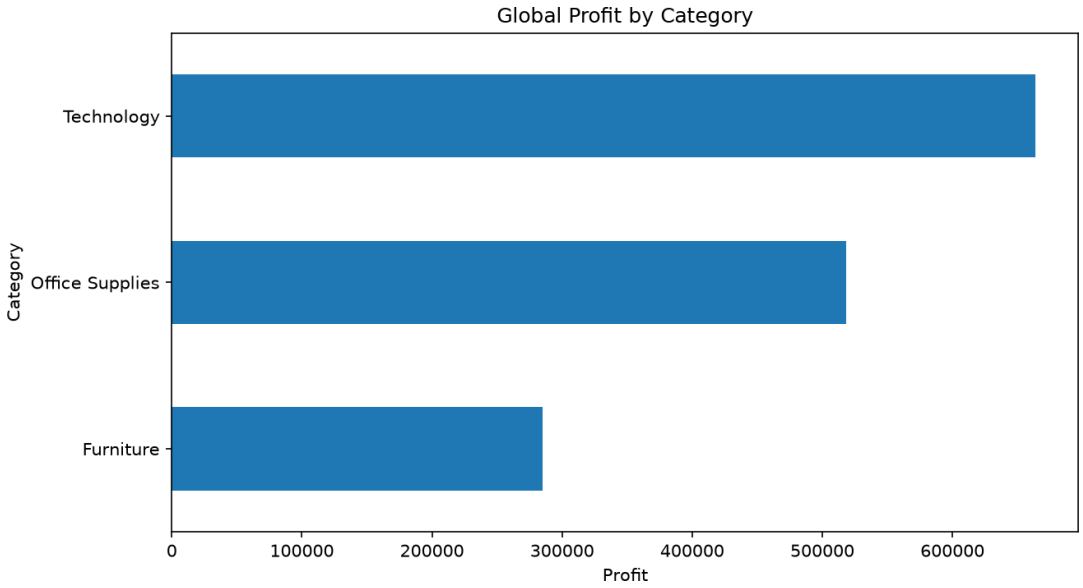


Figure 13.23. Dataset 2 profit by category.

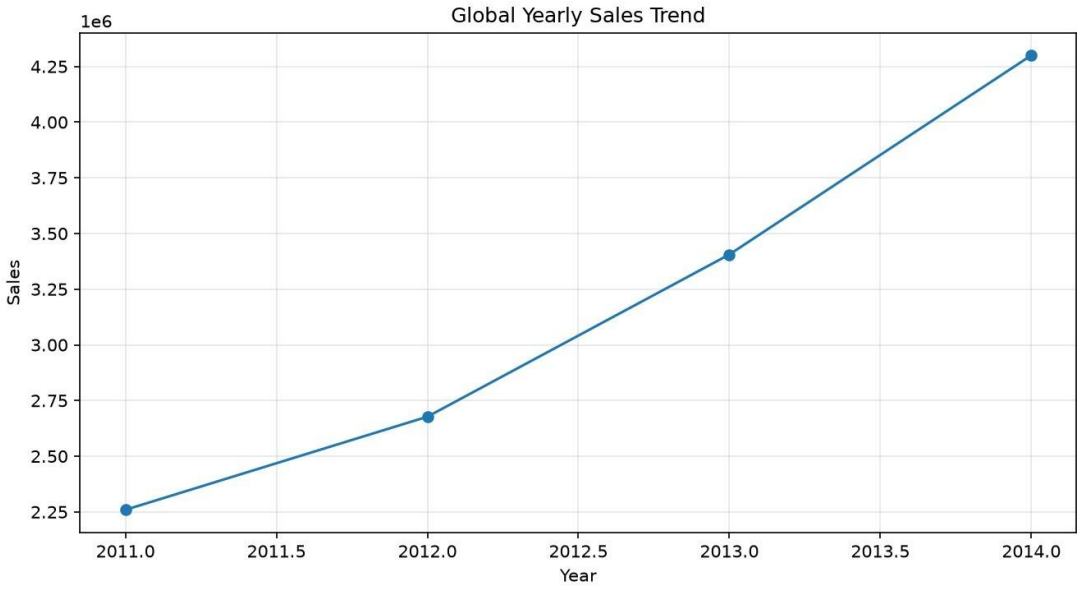


Figure 13.24. Dataset 2 yearly sales trend.

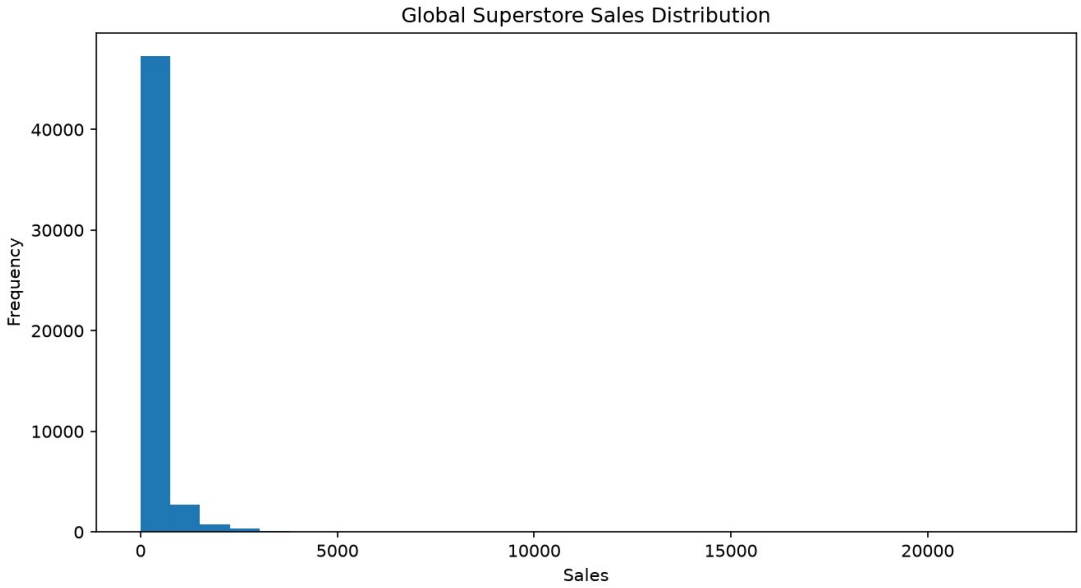


Figure 13.25. Dataset 2 sales distribution.

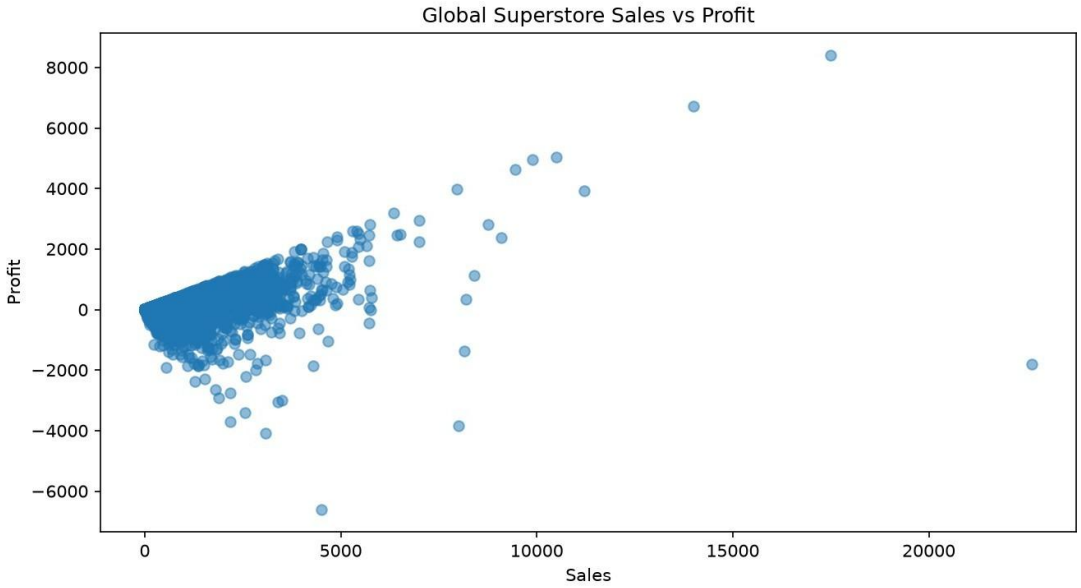


Figure 13.26. Dataset 2 sales versus profit.

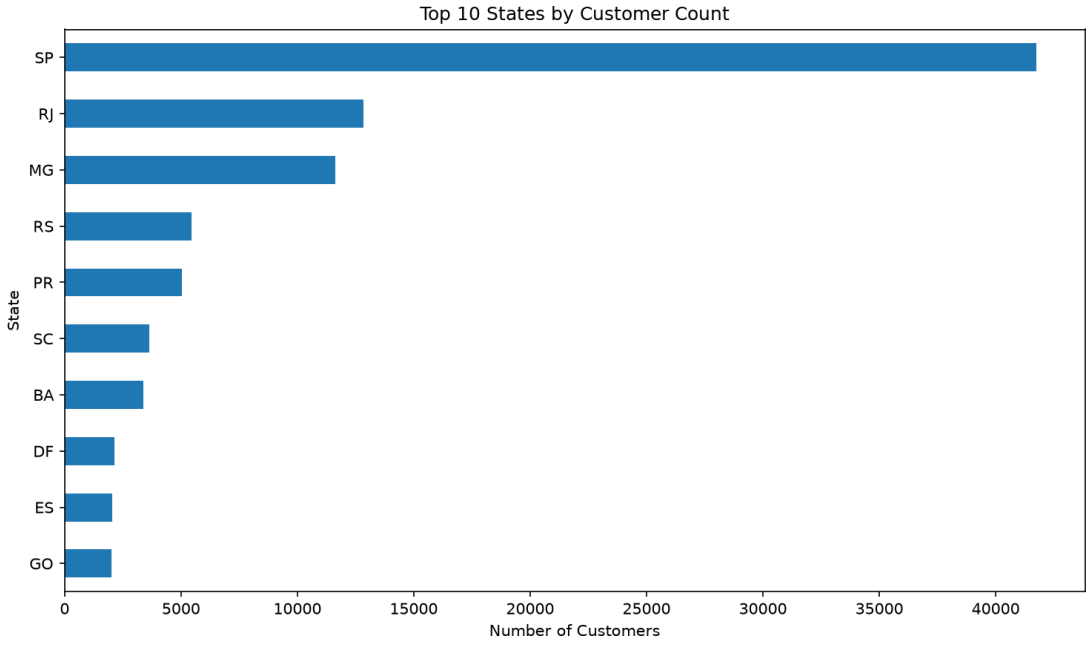


Figure 13.27. Olist customer records by state.

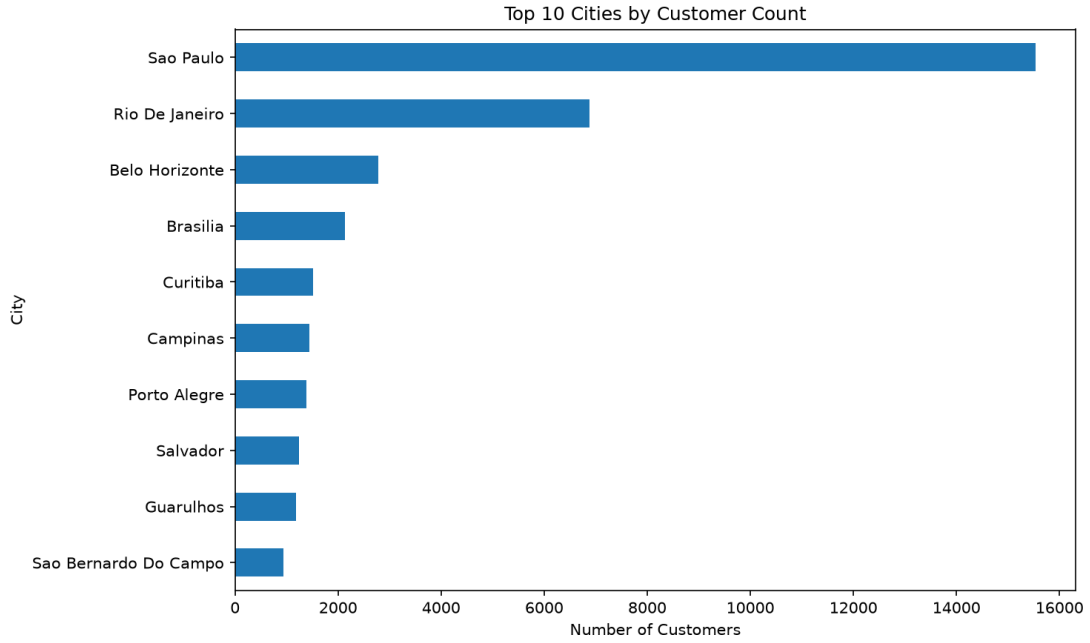


Figure 13.28. Olist customer records by city.

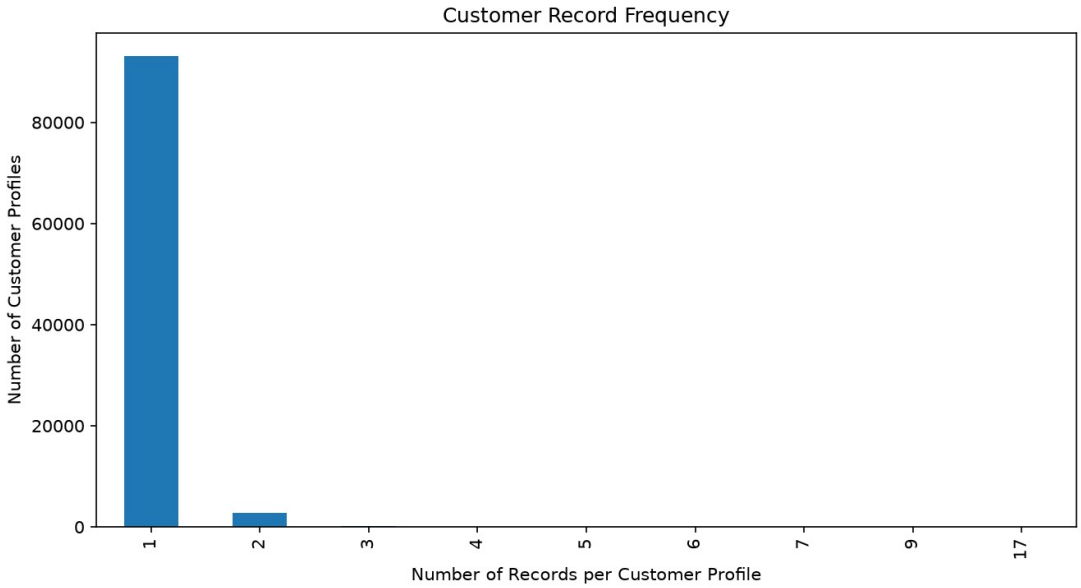


Figure 13.29. Olist customer record-frequency profile.

13.7 Business Summary and Decision-Support Outputs

Metric	Saved value	Description
Total Sales	2326534.3543	
Total Profit	292296.8145999999	
Total Quantity	38654.0	
Average Discount	0.1553845399254463	
Overall Profit Margin	12.563614805849074	

Category	Sales	Profit	Avg. discount	Profit margin
Technology	839,893.28	146,543.38	0.131	0.174
Office Supplies	731,893.31	126,023.44	0.156	0.172
Furniture	754,747.76	19,730.00	0.173	0.026

Region	Sales	Profit	Avg. discount
West	739,813.61	110,798.82	0.109
East	691,828.17	94,883.26	0.143
South	391,721.91	46,749.43	0.147
Central	503,170.67	39,865.31	0.241

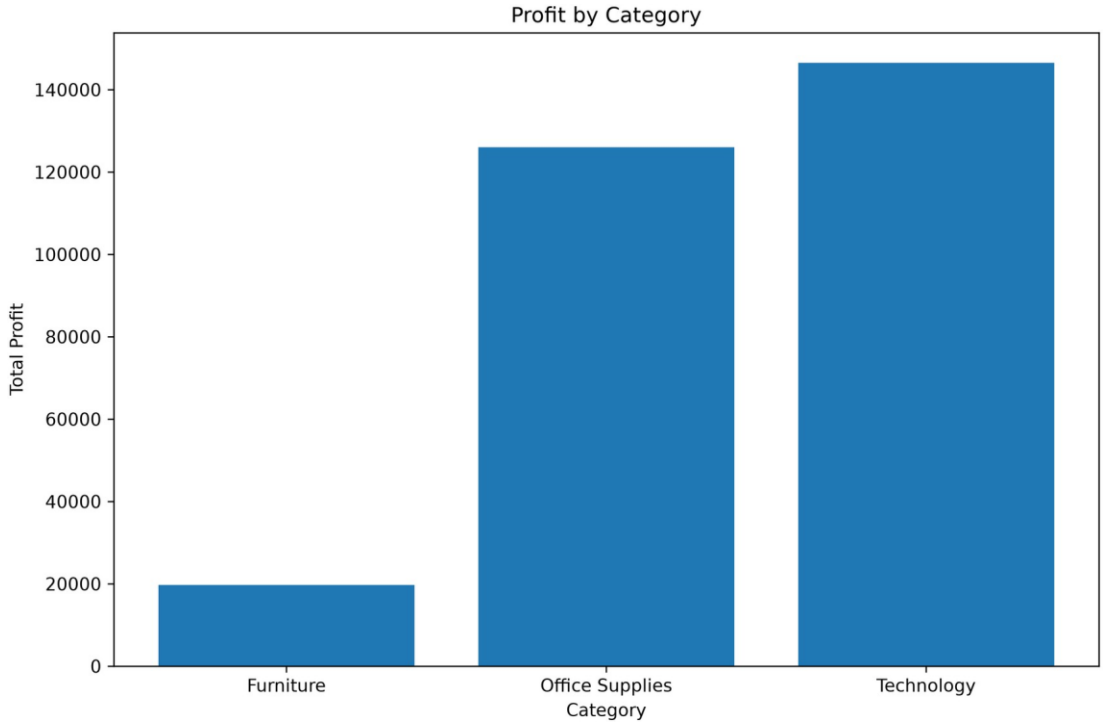


Figure 13.30. Business profit by category.

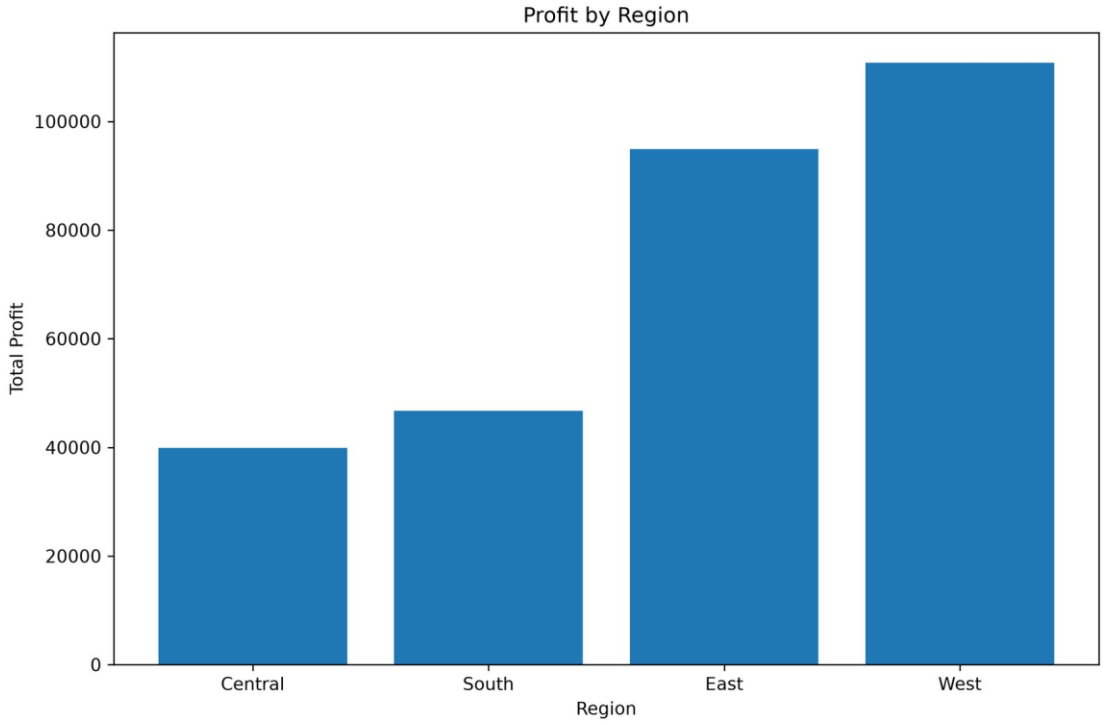


Figure 13.31. Business profit by region.

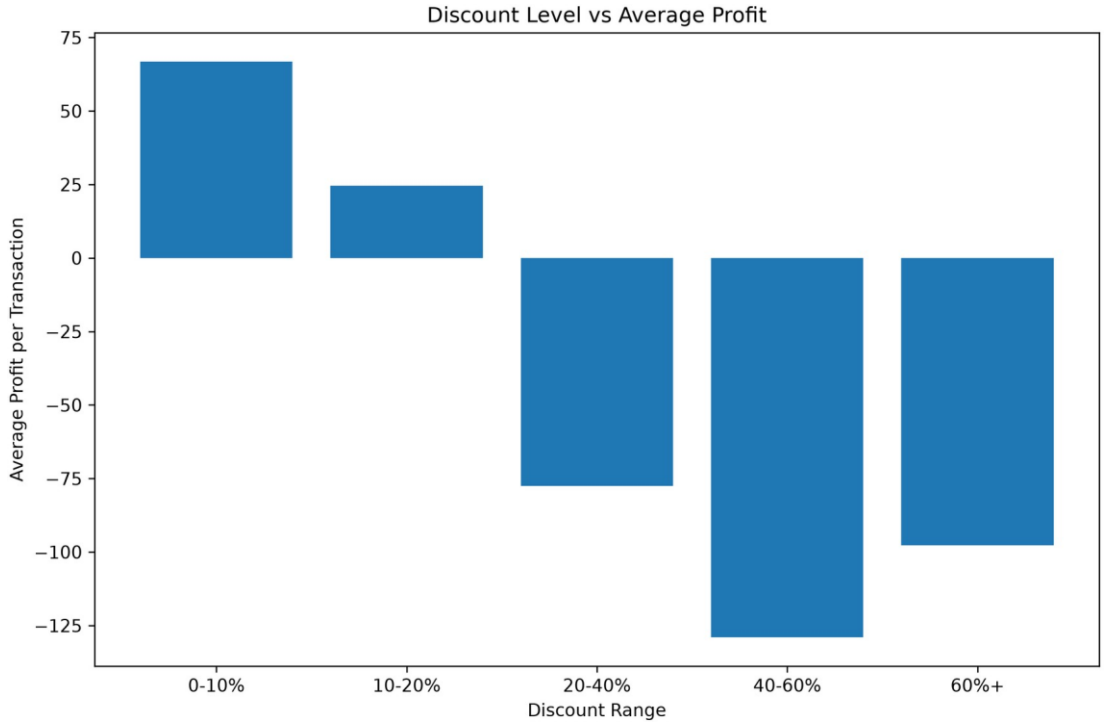


Figure 13.32. Discount versus profit output.

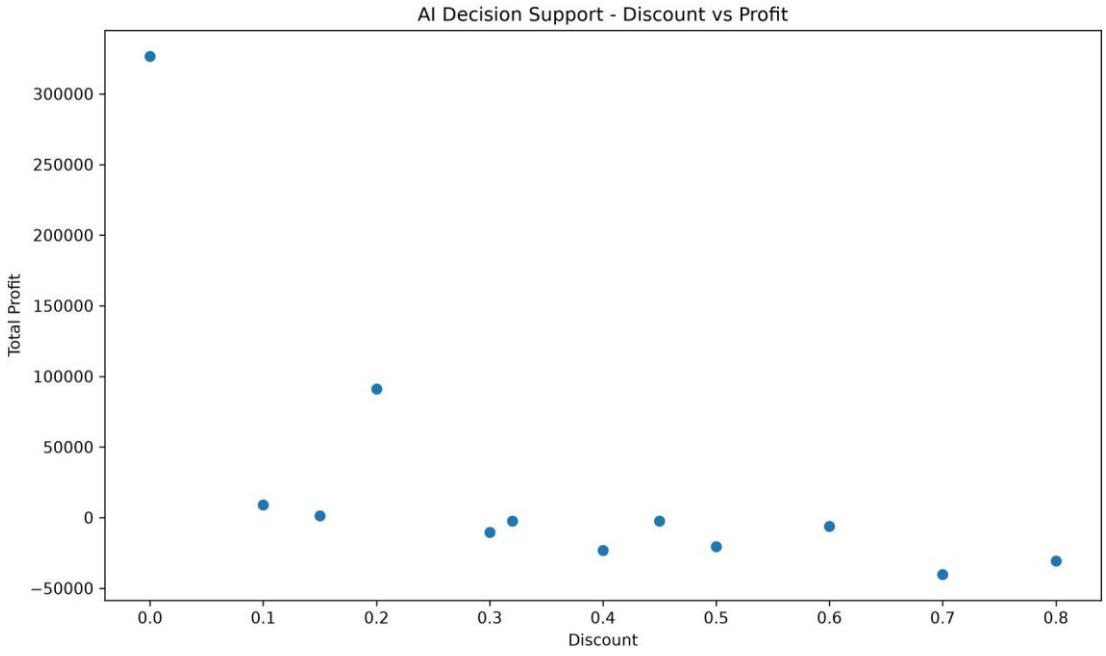


Figure 13.33. Decision-support discount/profit view.

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IMPORTANT FINALIZATION NOTE

This document is deliberately evidence-safe. It does not invent a final hybrid result that has not yet been executed. For the faculty submission, the remaining required action is to run the final hybrid experiment, insert the measured results into Sections 5–6, perform the final novelty check, add the final dashboard screenshots, complete plagiarism checking, and then update the repository and manuscript.